

**Goldbell Car Rental**

FLEET MANAGEMENT PLATFORM

Document Cost Savings Analysis

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Prepared by IT Department

Version 1.0 — Final

GBCR Platform — Cost Savings Analysis

In-House Development vs External Vendor

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1 Executive Summary

Goldbell Car Rental achieved exceptional cost savings by building the GBCR Fleet Management Platform entirely in-house, leveraging AI-assisted development tooling (Claude Code Pro) instead of engaging an external software vendor. An external provider quoted **SGD 50,000** for initial implementation plus **SGD 1,000 per month** in ongoing support — totalling **SGD 86,000 over three years**. The in-house approach delivered the same scope of work for approximately **SGD 3,240 per year** (the cost of the AI development tool alone), representing savings of **95% in Year 1** and **89% over three years**. Beyond pure cost reduction, this approach ensures full intellectual property ownership, zero vendor lock-in, and complete alignment with Goldbell's operational workflows — a strategic capability that can be replicated across future digital transformation initiatives.

2 Project Overview — What Was Built

Fleet Management Platform

Full-stack web application built with Next.js, React, and TypeScript. Production-grade architecture with server-side rendering and optimised performance.

Real-Time Vehicle Tracking & Booking

Live fleet status, availability calendars, booking workflows, and vehicle assignment — replacing manual spreadsheet-based processes.

Microsoft SSO & Azure AD Integration

Single sign-on via Azure Active Directory. Role-based access control for internal staff with no additional identity infrastructure required.

Enterprise System Integrations

Bidirectional data sync with Microsoft Dynamics AX (ERP), IBM Maximo (asset management), and Microsoft Fabric (analytics platform).

Customer Management & Inspection Workflows

Customer CRM, digital vehicle inspection checklists, condition reporting, defect tracking, and maintenance scheduling.

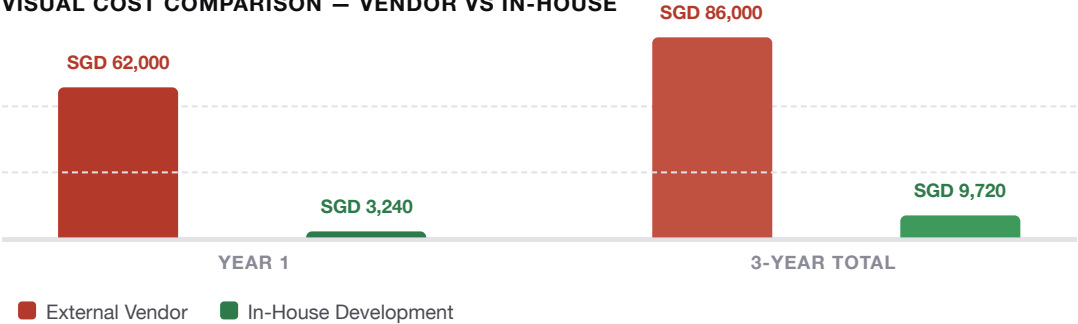
AI-Powered Analytics & PWA

Embedded analytics dashboards, AI-driven fleet insights, and a Progressive Web App shell for mobile-responsive access on any device.

3 Cost Comparison

● External Vendor	● In-House Development
Implementation (one-time)SGD 50,000	Claude Code Pro (~SGD 270/mo)SGD 3,240/yr
Support Year 1 (SGD 1,000 × 12)SGD 12,000	Azure AD / SSO SGD 0
Support Year 2 (SGD 1,000 × 12)SGD 12,000	MSSQL Database SGD 0
Support Year 3 (SGD 1,000 × 12)SGD 12,000	Microsoft Fabric SGD 0
Total (3 Years)SGD 86,000	Hosting (company servers)SGD 0
	Developer (on payroll)SGD 0
	Total (3 Years)~ SGD 9,720

VISUAL COST COMPARISON — VENDOR VS IN-HOUSE



4 Savings Summary

YEAR 1 TOTAL SAVINGS

SGD 58,760

95% reduction

Vendor Year 1: SGD 62,000 · In-House Year 1: SGD 3,240

3-YEAR CUMULATIVE SAVINGS

SGD 76,280

89% reduction

Vendor 3-Year: SGD 86,000 · In-House 3-Year: SGD 9,720

5 Strategic Advantages



Full IP Ownership

All source code, data schemas, and business logic belong entirely to Goldbell. Zero dependency on vendor licensing.



Same-Day Deployment

Features and bug fixes can be deployed immediately — no vendor SLA delays, change-request queues, or release windows.



Custom-Fit Workflows

Platform is built around Goldbell's exact processes. No forced adaptation to generic vendor software patterns.



Internal Knowledge Retention

Deep platform expertise stays within the IT team — no knowledge drain if the vendor relationship ends.



No Change-Request Fees

Scope changes, new features, and integration updates cost nothing beyond internal time — no vendor invoices.



Scalable to Other Units

The same development model — and platform components — can be extended to other Goldbell business units without additional vendor costs.

6 Risk Mitigation



Source code fully owned and version-controlled. The complete codebase is managed in a Git repository under Goldbell's control, with full commit history, branching strategy, and code review workflows in place.



Built on mainstream, widely-supported technology. Next.js, React, TypeScript, and Microsoft SQL Server are industry-standard technologies with large talent pools and long-term vendor support — not proprietary stacks.



Any competent developer can maintain the codebase. The platform is not dependent on a single individual. Standard patterns, documented APIs, and modern tooling ensure any qualified developer can contribute or take ownership.



Documentation and architecture are well-structured. Technical documentation, data models, API specifications, and deployment guides are maintained alongside the codebase to ensure continuity.

7 Limitations & Acknowledged Risks

While the in-house approach delivers significant cost and strategic advantages, the following limitations are acknowledged for full transparency and informed decision-making.

! **Key-person dependency.** The platform was primarily developed by a single internal resource. Although documentation and standard patterns reduce this risk, a transition period would be required if the developer were to leave. Mitigation: cross-training a second team member and maintaining up-to-date technical documentation.

! **Developer time not costed separately.** The cost comparison lists "Developer (on payroll) — SGD 0" because the developer is an existing staff member. The actual time invested in building and maintaining the platform represents an opportunity cost that is not captured in the figures above.

! **Ongoing maintenance is an internal responsibility.** Unlike a vendor engagement that includes contracted support and SLAs, all future maintenance — security patches, dependency upgrades, framework updates, and bug fixes — falls on the internal team indefinitely.

! **No contractual SLA for support response.** A vendor typically guarantees response and resolution times. The in-house model relies on internal resource availability, which may be affected by leave, competing priorities, or team capacity.

! **AI tooling dependency.** Development velocity relies on continued access to AI-assisted development tooling (Claude Code Pro). Changes in pricing, availability, or capability of this tool could affect future development speed and cost assumptions.

! **No independent security audit or load testing.** The platform has not yet undergone a third-party security penetration test or formal load/stress testing. A vendor engagement would typically include these as part of the delivery. Recommended: schedule an independent security review before expanding the platform's user base.

8 Recommendation

Based on the demonstrated cost savings of **89–95%**, full IP ownership, rapid delivery, and strategic alignment with Goldbell's operational needs, the IT Department strongly recommends **continuing the in-house, AI-assisted development approach for all future digital projects** where scope and complexity are well-understood. This model — leveraging existing infrastructure, existing staff, and AI development tooling — has proven to deliver enterprise-grade software at a fraction of vendor cost while retaining control, flexibility, and institutional knowledge within the organisation. Where external expertise is genuinely required, it should be engaged selectively for narrow, defined deliverables rather than broad platform ownership.

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